

Elias Dengo

Software Engineer

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Professional Summary

Results-driven Software Engineer with over 2 years of experience in designing, developing, and maintaining scalable, secure, and high-performance software solutions. Proven expertise in full-stack development, API integration, database architecture, and system optimization, with a strong track record of delivering business-aligned software in the financial sector. Adept at translating business requirements into technical solutions, collaborating with cross-functional teams, and driving continuous improvement. Seeking to contribute to Habesha Breweries S.C. by developing reliable, efficient, and user-friendly software that supports operational excellence and future-proof production.

Education

B.Sc. in Software Engineering

Arba Minch University | Graduated July 2023 | GPA: 3.90

Certifications

Oracle SQL Database – Learn Quest (2023)

System Administration and IT Infrastructure Services – Google (2023)

Python Data Structures – University of Michigan (2022)

Project Management Fundamentals – Google (2023)

Technical Skills

Programming Languages: Python, Java, JavaScript, PHP, C, C++

Frameworks & Technologies: Django, Spring Boot, Node.js, Express.js, RESTful APIs, Web Services

Databases: MySQL, PostgreSQL, Oracle, MS SQL Server, MongoDB

Tools & Platforms: Git, Linux, Docker, CI/CD, WSO2 API Manager, ERP Next

Cloud & DevOps: System Administration, IT Infrastructure, Networking, Performance Tuning

Specialized Skills: API Development & Integration, Database Design & Optimization, ETL Processes, System Architecture, Test Automation

Work Experience

Database Administrator and Integration specialist

Zemen Bank S.C. | Sep 2023 – Present

Designed, implemented, and maintained secure and scalable database architectures for critical financial systems.

Developed and optimized complex SQL queries and ETL processes to support business intelligence and reporting.

Ensured high availability and performance through proactive monitoring, backup strategies, and disaster recovery plans.

Collaborated with cross-functional teams to integrate systems using Service-Oriented Architecture (SOA).

Senior Software Engineer (Multi-role)

GTAP Payment Gateway (Freelancer)

Managed CI/CD pipeline and infrastructure automation for system deployment.

Developed RESTful APIs for integration with third-party services.

Optimized databases for performance and implemented security measures.

Software Engineer

Wegagen Bank S.C. | Jun 2022 – Nov 2022

Developed and maintained banking applications using Java, Spring Boot, and web technologies.

Conducted testing, debugging, and performance tuning of financial software systems.

Collaborated with stakeholders to gather requirements and deliver user-centric solutions.

Software Engineer Intern

Ministry of Innovation and Technology | Jun 2022 – Nov 2022

Assisted in the development and deployment of software solutions for innovative technology projects.

Participated in code reviews, quality assurance, and technical documentation.

Freelance Senior Software Engineer

GTAP Payment Gateway | Project-based

Built and managed CI/CD pipelines for automated deployment and system scaling.

Developed secure RESTful APIs and integrated third-party payment and IOT services.

Optimized database performance and implemented robust security protocols.

Key Projects

FinTech Machine learning Project

This fintech project deploys multiple predictive systems for car pricing, fraud detection, house valuation, and loan eligibility into a full-stack application. It uses a FastAPI backend with scikit-learn and XGBoost

models, serialized via Joblib, and a modern Next.js/Typescript frontend. The architecture handles structured financial data, focusing on both regression and classification tasks for real-world decision-making. Key challenges included managing imbalanced fraud data and ensuring low-latency predictions for user applications. The system demonstrates a scalable pipeline for automating and augmenting financial analysis and risk assessment.

Health Machine learning Project

This project deploys multiple health prediction models (heart disease, COVID-19, diabetes, calories) into a production web application. It features an ML backend built with FastAPI and a responsive frontend using Next.js. The system demonstrates a complete pipeline from data processing and model training to serialization and real-time API serving. It tackles challenges like imbalanced medical data

Loan Request Management System

Led end-to-end development of a full-stack system using Node.js, Express.js, and MongoDB.

Designed RESTful APIs with JWT authentication and role-based access control.

Integrated with legacy HR and Finance systems (MS SQL Server) via secure APIs.

Reduced loan processing time by 90% through automated multi-level approval workflows.

Church ERP System (ERP Next-based)

Architected and developed a custom ERP system with modules for Finance, Payroll, Inventory, and more.

Implemented PostgreSQL and MS SQL Server integration for seamless data flow.

Automated manual processes, reducing administrative workload by over 70%.

Wegagen Bank Link Management System

Developed a centralized PHP/MySQL platform to streamline internal application access.

Applied MVC architecture to improve usability and reduce IT support calls.

E-commerce & Delivery System

Built a scalable backend using Node.js, Express.js, and MySQL with real-time Web Socket tracking.

Integrated payment gateways (Stripe, PayPal) and third-party services.

Extracurricular & Leadership

Sub-Committee Officer, Arba Minch University Student Peace Forum (2020–2023)

Volunteer Teacher, Teaching IT and Mathematics to high school students (2021)

Software Engineer, Arba Minch Science and Technology Club (2020–2023)